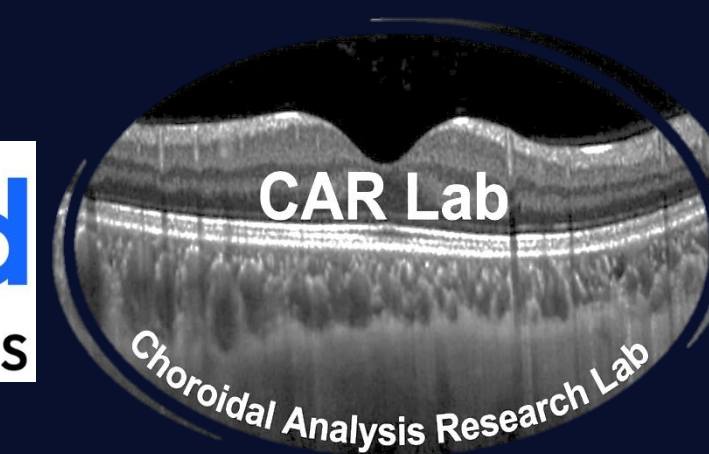




The Disconnect Between Device Quality Score and Deep Learning based Choroid Segmentation in Retinal OCT

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INTRODUCTION

- Device-generated Image Quality (IQ) > 15 thresholds are widely used as quality gate for clinical studies.
- This study assesses whether IQ meaningfully correlates with deep learning performance, measured by segmentation accuracy and choroid volume quantification.

METHODS

- Dataset:** OCT macular scans: 63 healthy subjects (103 volumes) Acquired on a Heidelberg Engineering Spectralis® device using 6 × 6 mm protocol; 496 × 512 × 49 acquisition grid; IQ: 27.65 dB ± 3.86 dB.
- Segmentation paradigms:**
 - Deep Learning:** NMI-ChoroidAI, an open-source deep learning model (DGPET)
 - Image processing:** NMI-ChoroidIP
- Quality-performance analysis:** Pearson correlation between
 - IQ scores per B-scan and segmentation accuracy (Dice coefficient).
 - Aggregated IQ scores across B-scans within each volume using moment-based (mean, standard deviation) and order-based (minimum, maximum, median, mode, range) statistics and choroid volume error (eChV).

CONCLUSIONS

- Across segmentation paradigms, errors occur largely independent of input image quality, indicating that device-generated IQ scores are insufficient to predict AI performance and that AI-specific reliability metrics are needed, particularly for quantitative, volume-based outcomes.

REFERENCES

- Burke, J., et. al (2023). An open-source deep learning algorithm for efficient and fully automatic analysis of the choroid in optical coherence tomography. *Translational Vision Science & Technology*, 12(11), 27-27.

RESULTS

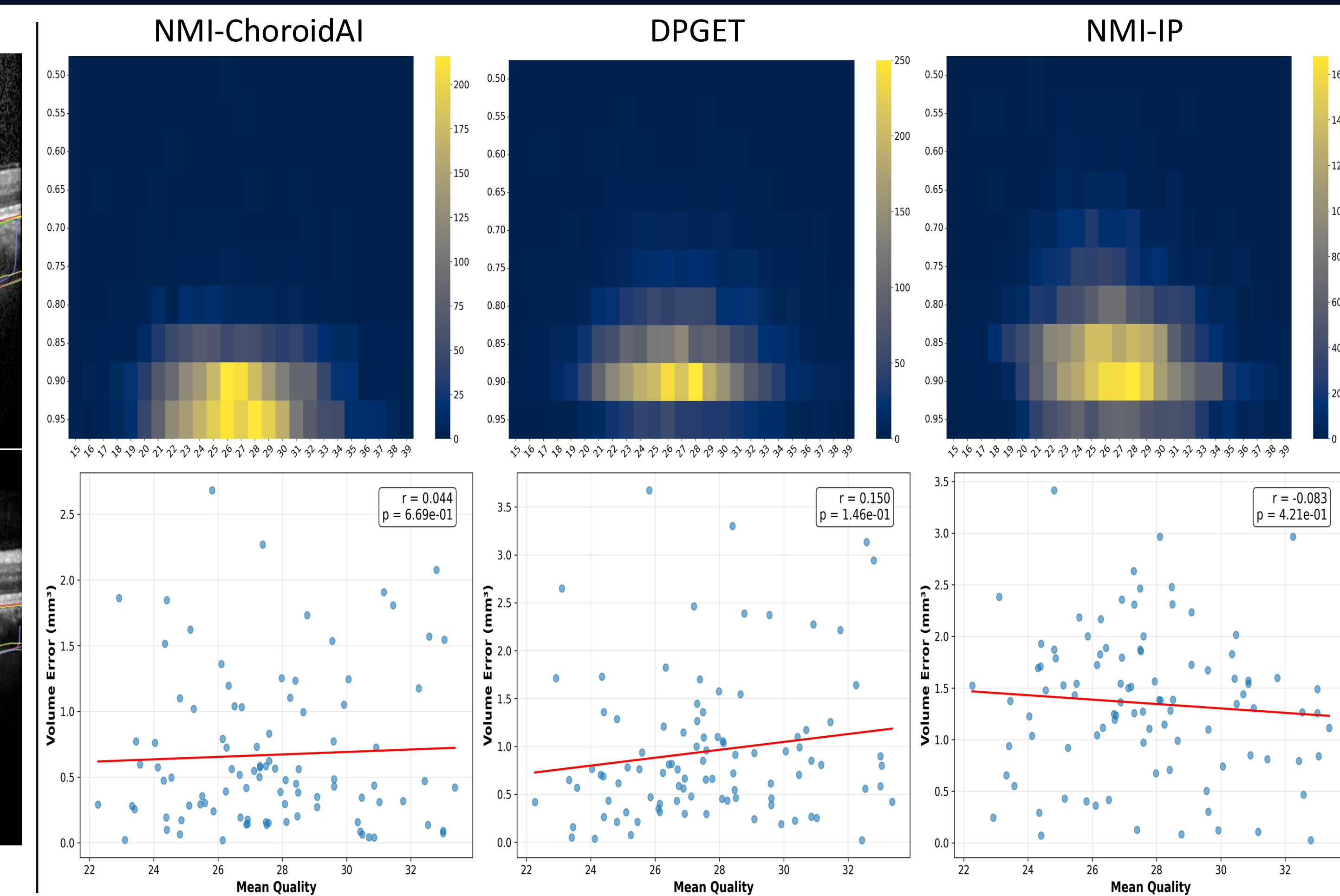
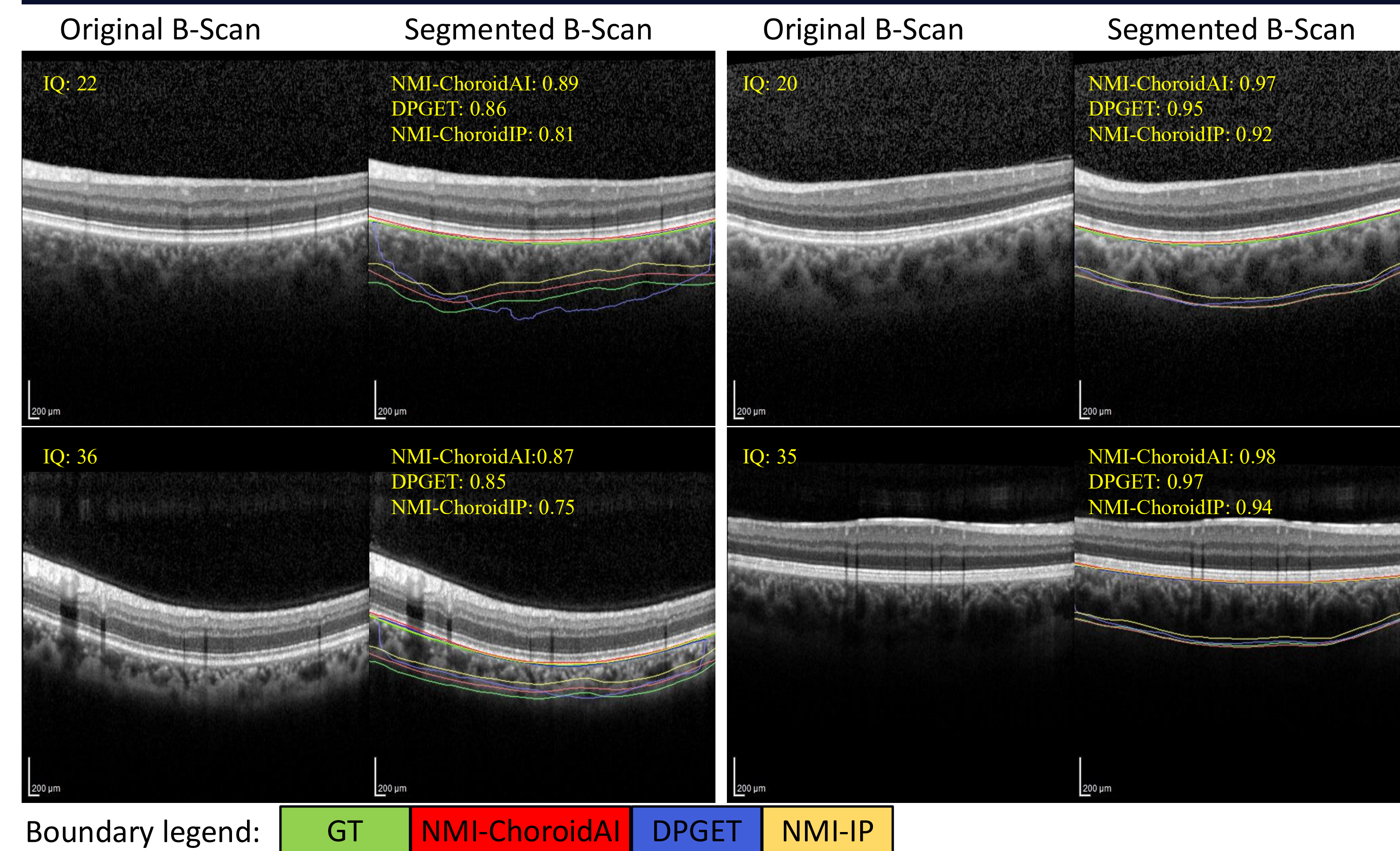


Figure: (Left) Representative examples of choroid segmentation illustrating combinations of Image Quality (IQ) and Dice coefficient. (Right-top) Relationship between IQ (x-axis) and segmentation accuracy measured by Dice coefficient (y-axis) for each model; color scale indicates co-occurrence of IQ and Dice values. Panels correspond to NMI-ChoroidAI (left), DGPET (middle), and NMI-ChoroidIP (right). (Right-bottom) Correlation between mean IQ aggregated across B-scans within each volume and choroid volume error (eChV).

- Segmentation performance:** Mean Dice coefficients were 0.94 ± 0.04 for NMI-ChoroidAI, 0.90 ± 0.05 for DGPET, and 0.89 ± 0.07 for the NMI-ChoroidIP model.
- IQ-performance correlation:** Pearson correlation coefficients between Image Quality (IQ) and segmentation accuracy were 0.055 (NMI-ChoroidAI), 0.005 (DGPET), and 0.122 (NMI-ChoroidIP), indicating negligible associations (large n; p-values not emphasized).
- Choroid volume error:** No meaningful relationship was observed between IQ metrics and choroid volume quantification error (eChV) across models. While NMI-IP has the highest mean eChV (1.35 cu. mm) consistent with the low DC, a weak association was noted between maximum IQ and eChV in the DGPET model ($r = 0.238$, $p = 0.020$).

IQ Metric	Pearson Correlation (p-value)		
	NMI-ChoroidAI	DGPET	NMI-ChoroidIP
Mean	0.044 (0.67)	0.150 (0.15)	-0.083 (0.42)
Std. dev.	-0.003 (0.98)	0.157 (0.13)	-0.141 (0.17)
Median	0.017 (0.87)	0.131 (0.20)	-0.074 (0.47)
Mode	0.064 (0.54)	0.153 (0.14)	-0.101 (0.33)
Min.	0.054 (0.60)	0.016 (0.88)	0.056 (0.59)
Max.	0.097 (0.35)	0.238 (0.02)	-0.083 (0.42)
Range	0.040 (0.70)	0.214 (0.04)	-0.135 (0.19)

Table: Pearson correlation values (p-value, n=103) between the Image Quality (IQ) and the error in choroid Volume across three different segmentation models



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